

Reconfigurable DDR5 Memory Controller

Architecture Planning Session — Analysis Report

Project Status · Architecture Review · Critical Issues · ADR Register · Open Issues · Obsidian Graph · Roadmap

Standard	JEDEC JESD79-5, DFI 5.2, AXI4	Interface	AXI4 → CIF → CDC → MC Core → DFI → PHY	Status	Arch
Overall	≈40% complete	RTL Ready	No	Freeze	Pen

1 System Architecture Overview

1.1 Top-Level Signal Flow

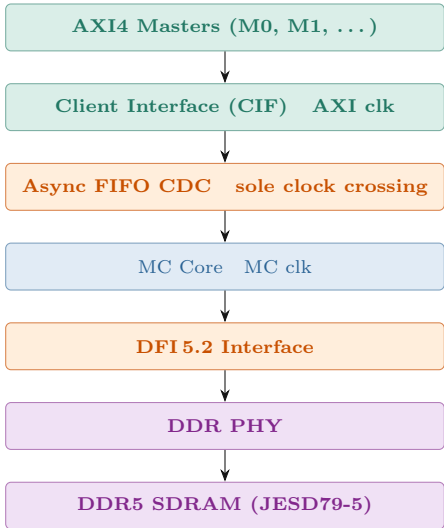


Figure 1. Full system hierarchy. Teal = CIF/AXI. Blue = MC Core. Amber = clock crossing / DFI boundary. Purple = PHY + DRAM.

1.2 System Scope Decisions (ADR-0000)

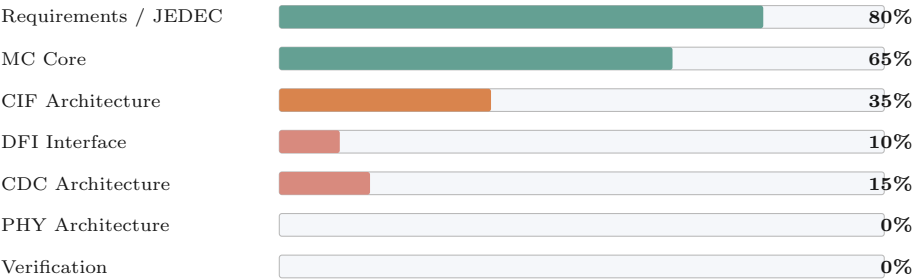
Parameter	Type	Value / Options
DDR Generation	Compile-time	DDR5 (JESD79-5)
Device Width	Compile-time	x4 / x8 / x16 — OPEN: ISSUE-0001
Density	Compile-time	8 Gb / 16 Gb / 32 Gb / 64 Gb — OPEN
Rank Count	Compile-time	Single / Multi — OPEN: ISSUE-0002
Topology	Compile-time	Discrete / UDIMM / SODIMM — OPEN
DFI Version	Compile-time	DFI 5.2
Page Policy	Runtime	Open / Closed / Adaptive
Refresh Policy	Runtime	REFab / REFsb / FGR — OPEN: ISSUE-0003
Timing Values	Runtime	All tXX from Config Registers
RFM Parameters	Runtime	RAAIMT, RAAMMT, RAADec — OPEN: ISSUE-0004
DFI Freq Ratio	Runtime	1:1, 1:2, 1:4 — OPEN: ISSUE-0005

2 Architecture Completeness

2.1 Per-Domain Completion

Domain	%	Status	Biggest Gap
Requirements / JEDEC	80%	Good	Bank organisation vs device width
CIF Architecture	35%	Partial	Command format, CDC contract, queue architecture
CDC Architecture	15%	Weak	Depth, backpressure, reset, overflow
MC Core	65%	Solid	Rank Manager, Maintenance Arbiter, DFI ownership
DFI Interface	10%	Weak	All six sub-interfaces unspecified
PHY Architecture	0%	None	Everything
Verification Architecture	0%	None	Everything
Overall	≈40%	Architecture Phase	—

2.2 Progress Bars



3 What Is Defined

3.1 MC Core — Defined Blocks

Block	Status	Coverage
Config Registers	Defined	Full register table (32+ regs), all timing params
Init FSM	Defined	16-state sequence, MRW burst, ZQCAL, DLL reset
Bank State Machines	Defined	11 states, 16 instances, all per-bank timers
Timing Control	Defined	All cross-bank constraints, tFAW window, countdown timers
Arbitration Logic	Defined	6-level priority, age counters, watermark switching
Command Selector	Defined	Page policy, BG-aware reorder, 16-gate legal check matrix
Refresh Scheduler	Defined	REFab, REFsb, FGR, leaky-bucket, urgent override
ZQcal FSM	Defined	MPC ZQCAL Start/Latch flow, periodic interval
RFM FSM	Defined	Per-bank RAA counters, RAAIMT threshold, tRFM
Power Management	Defined	Precharge PD, Active PD, Self-Refresh, MPSM
Write Data Path	Defined	BL16 buffer, DM/ECC, Write CRC, WL align, DFI timing
Read Data Path	Defined	Latency counter, data FIFO, ECC/CRC checker, ECS FSM

3.2 CIF — Defined Blocks

Block	Status	Coverage
AXI Write Port	Partial	AW/W channel handlers, B-response generator
AXI Read Port	Partial	AR channel, outstanding request tracking
Burst Splitter	Partial	Page boundary split (Stage 1), BL alignment (Stage 2)
Address Translator	Partial	Bit-field extractor, CMD builder, configurable mapping
Write Cmd Pool	Partial	Entry storage, allocation, dispatch identified
Read Cmd Pool	Partial	Entry storage, allocation, dispatch identified
Reorder Buffer (ROB)	Partial	Per-AXID queues, seqnum tagging, HEAD retirement
Merge Logic	Partial	Fragment buffer, reassembly tracker, output mux
Response Tracking	Partial	Seqnum generator, tag insertion, tag lookup

4 Critical Issues

CRITICAL-1 — Bank Organisation Inconsistency

Documents assume **16 banks** (**2 BG** × **4 banks** × **2 sub-channels**).

DDR5 actual organisation:

- x4/x8 devices: 8 BG × 4 banks = **32 banks**
- x16 devices: 4 BG × 4 banks = **16 banks**

Impact: Address Mapper, Bank FSM count, Refresh Scheduler, all timing formulae.

Resolution: Answer ISSUE-0001 (target device width + density) before any RTL.

CRITICAL-2 — Missing Rank Manager

Rank state is currently scattered across Refresh Scheduler, Power FSM, and Timing Control.

Required dedicated block owning: `rank_busy`, `rank_refresh`, `rank.power.state`, `rank.self.refresh`, `rank_zq_block`, `rank_cke`

Impact: DFI integration will be painful without a single rank state source of truth.

CRITICAL-3 — Missing Maintenance Arbiter

REF, RFM, ZQ, MRR, and Scrubber run independently with no defined collision behaviour.

Scenario: `ref_urgent` + `rfm_trigger` + `zq_expiry` same cycle.

Required: **Maintenance Arbiter** with explicit priority and mutual-exclusion rules.

Required document: `Maintenance_Arbiter.md`

CRITICAL-4 — DFI Training Ownership Undefined

TRAINING_DISPATCH exists inside Init FSM but ownership boundary is undefined.

Required separation:

- MC owns: sequencing, handshake signals, retry logic
- PHY owns: execution, measurement, result reporting
- DFI owns: handshake protocol (`dfi_wrlvl.en/resp`, `dfi_rdlvl.en/resp`)

Required document: `DFI_Training_Orchestrator.md`

ARCH-1 — Duplicate Queue Structure Risk

CIF contains Write Cmd Pool and Read Cmd Pool.

MC Core also referenced RD/WR pools downstream of CDC.

Option A (correct): CIF Pools → CDC → MC Scheduler (no duplicate queuing)

Option B (incorrect): CIF Pools → CDC → MC Pools → Scheduler (extra latency, extra storage)

Resolution required in ADR-0004.

ARCH-2 — Address Mapper has No Owner

Physical address → Channel → Rank → BG → Bank → Row → Column mapping is referenced in CIF and Config Registers but owned by neither.

Required: explicit **Address Mapper** block between Burst Splitter and Cmd Pools.

Required document: `Address_Mapping.md`

ARCH-3 — Missing Command Queue Architecture

Command entry format, depth, allocation, retirement, hazards, reordering limits, and age counter details are unspecified.

Required document: `Command_Queue_Architecture.md`

5 What Is Missing**5.1 MC Core — Missing Blocks**

Block	Priority	Reason Needed
Rank Manager	P0 — Critical	Single source of truth for all rank state; required before DFI integration
Maintenance Arbiter	P0 — Critical	Collision arbitration for REF/RFM/ZQ/MRR/Scrub
DFI Training Orchestrator	P0 — Critical	Training ownership boundary between MC, DFI, PHY
Address Mapper	P0 — Critical	Physical → {channel, rank, BG, bank, row, col}
Command Queue Spec	P1 — High	Entry format, depth, hazards, retirement rules
CDC Contract	P1 — High	FIFO depth, backpressure, overflow, underflow, reset
Statistics Block	P2	row_hits, misses, BG conflicts, read/write latency
Performance Monitor	P2	Cmds issued, refresh count, stall cycles, bank utilisation
Debug Architecture	P2	Trace buffer, event log, status registers
Error Architecture	P2	ECC, CRC, training failures, refresh violations
Interrupt Controller	P2	init_done, training_done, ecc_error, fatal_error

5.2 CIF — Missing Documents

Document	Blocks It Resolves
<code>CIF_Architecture.pdf</code>	Full CIF decomposition and interfaces
<code>Address_Mapping.pdf</code>	Field extraction, configurable bit-mapping, DDR3/4/5 support
<code>Burst_Splitter.pdf</code>	Stage 1 (page boundary), Stage 2 (BL alignment), math
<code>Write_Cmd_Pool.pdf</code>	Entry format, depth, allocation, dispatch, free
<code>Read_Cmd_Pool.pdf</code>	Same as write
<code>Response_Tracking.pdf</code>	Seqnum generator, tag format, lookup
<code>Reorder_Buffer.pdf</code>	Per-AXID queues, HEAD retirement, isolation guarantees
<code>Merge_Logic.pdf</code>	Fragment buffer, reassembly, bypass path

5.3 DFI Interface — Missing Sub-Interfaces

Sub-Interface	Status
Command Interface	Partially mapped (address/bank/bg/cs signals in MC doc)
Read Path	rddata, rddata_valid, rddata_en, rddata_dnv
Write Path	wrdata, wrdata_mask, wrdata_en, wrdata_cs_n, parity
Update Interface	ctrlupd_req/ack, phyupd_req/ack — undefined ownership
Training Interface	wrlvl_en/strobe, rdlvl_en/resp, calvl_en — undefined
Low Power Interface	cke, cksre/srx, init_start/complete — partially mapped

6 Obsidian Knowledge Graph

6.1 Root Graph Structure

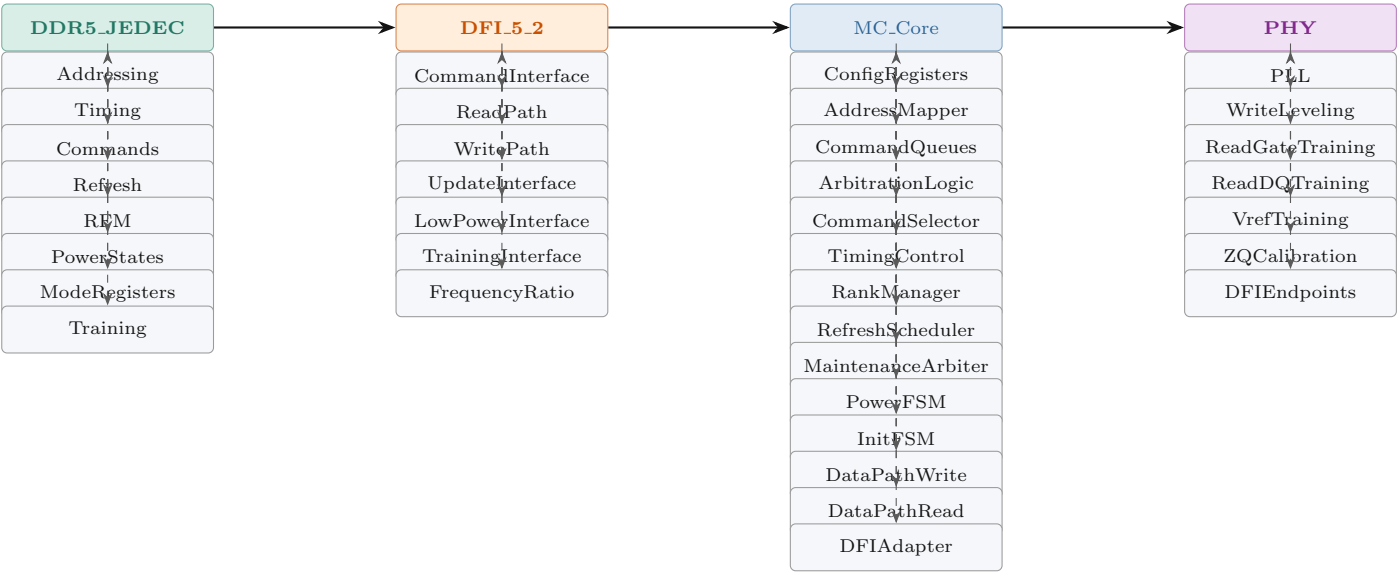


Figure 2. Obsidian root knowledge graph. Each column is a top-level node; children are sub-topic notes. Arrows show dependency direction (JEDEC → DFI → MC → PHY).

6.2 CIF Sub-Graph

CIF Node	Child Nodes
AXI.Write_Port	AW_Channel_Handler, W_Channel_Handler, Write_Request_Validation, B_Response_Generator
AXI.Read_Port	AR_Channel_Handler, Read_Request_Validation, Outstanding_Request_Tracking, Read_Request_Formatter
Burst_Splitter	Page_Boundary_Detector, Burst_Split_Logic, Command_Generator, Metadata_Generator
Address_Translator	Address_Mapping, Field_Extractor, Mapping_Config, Address_Validation
Write_Cmd.Pool	Entry_Storage, Allocation, Dispatch, Free
Read_Cmd.Pool	Entry_Storage, Allocation, Dispatch, Free
Response_Tracking	Sequence_Number_Generator, Tag_Generator, Attribute_Buffer, Tag_Lookup
Reorder_Buffer	Per_AXID_Queue, Fill, Ordering, Retirement
Merge_Logic	Response_Collection, Burst_Reconstruction, Response_Merge, AXI_Response_Formatter

7 Architecture Decision Records (ADR)

7.1 Completed ADRs

ADR	Title	Decision
ADR-0000	System Scope	Compile-time: DDR gen, width, density, rank, DFI version. Runtime: policy, timing, RFM.
ADR-0000A	Config Model	Boot-time configurable vs runtime. Requires-reset and requires-retraining flags defined.
ADR-0000B	Ownership Matrix	All 14 MC Core blocks + CIF blocks assigned explicit ownership.
ADR-0000C	Context Objects	Bank Context, Rank Context, Command Context, Refresh Context all defined.

7.2 Required ADRs — Priority 0

ADR	Title	Blocks on This
ADR-0001	Command Format	Queue depth, entry fields, tag format, allocation, retirement
ADR-0002	Address Mapping	Physical → {channel, rank, BG, bank, row, col}; configurable bit-field scheme
ADR-0003	CDC Contract	FIFO depth, backpressure protocol, overflow/underflow, reset strategy
ADR-0004	Queue Architecture	CIF pool vs MC pool decision (Option A vs B); hazard rules; reordering limits
ADR-0005	Command Lifecycle	Allocation → dispatch → issue → completion → retirement state machine
ADR-0006	Timing Ownership	Which block owns which constraint timer; cross-block update protocol

7.3 Required ADRs — Priority 1

ADR	Title	Blocks on This
ADR-0007	Rank Manager	rank_busy, rank_refresh, rank_power, rank_sr, rank_zq — all signals
ADR-0008	Maintenance Arbiter	Priority order for REF/RFM/ZQ/MRR/Scrub collision resolution
ADR-0009	Training Architecture	MC/DFI/PHY boundary; write leveling; read gate; DQ; VREF; CA; CS
ADR-0010	DFI Ownership	Which MC sub-block drives which DFI signal; timing adapters

7.4 Required ADRs — Priority 2

ADR	Title
ADR-0011	PHY Architecture (PLL, clocking, read/write paths, calibration)
ADR-0012	Verification Architecture (coverage model, assertions, scoreboards, compliance)

8 Open Issues Register

ID	Title	Status	Impacts
ISSUE-0001	Target DRAM Organisation	OPEN	Bank count, Address Mapper, BSM instances, Refresh Scheduler
ISSUE-0002	Single vs Multi Rank	OPEN	Refresh, Power Management, Training, DFI per-rank signals
ISSUE-0003	Refresh Strategy	OPEN	REFab vs REFsb vs hybrid; FGR mode; tRFCsb in BSM
ISSUE-0004	RFM Support Level	OPEN	Mandatory/optional/disabled; RAAIMT/RAAMMT register fields
ISSUE-0005	DFI Frequency Ratio	OPEN	1:1/1:2/1:4; all DFI timing formulae depend on this
ISSUE-0006	PHY Ownership Boundary	OPEN	Which signals MC drives vs PHY drives at DFI boundary
ISSUE-0007	CIF Duplicate Queue Risk	OPEN	Option A vs B; resolves in ADR-0004

9 Context Objects

Context	Fields
Bank Context	bank_state, open_row, active_row, tRCD, tRAS, tRP, tRFC
Rank Context	rank_state, refresh_state, power_state, zq_state
Command Context	command_id, axi_id, rank, bg, bank, row, column, priority, age, timestamp
Refresh Context	credits, last_refresh, urgent_flag, mode (REFab/REFsb/FGR)

10 Ownership Matrix

Block	Layer	State	Interfaces
AXI Write/Read Port	CIF	Partial	AXI AW/W/AR/B/R channels
Burst Splitter	CIF	Partial	AXI burst → sub-transactions
Address Translator	CIF	Partial	byte_addr → {rank, BG, bank, row, col}
Cmd Pools (WR/RD)	CIF	Partial	Accepts from translator; drives CDC
Reorder Buffer	CIF	Partial	Tags, per-AXID queue, HEAD retirement
Merge Logic	CIF	Partial	Fragment reassembly, ROB feed
Async FIFO CDC	CDC	Weak	AXI clk → MC clk crossing
Config Registers	MC	Defined	APB/CSR slave, all timing params
Init FSM	MC	Defined	DFI direct; INIT_KICK → init_done
Bank State Machines	MC	Defined	16 instances; per-bank timers
Timing Control	MC	Defined	cmd_ok[15:0][cmd_type] per cycle
Arbitration Logic	MC	Defined	Priority, age, watermark; winner class
Command Selector	MC	Defined	Legal check, page policy, sequence gen
Refresh Scheduler	MC	Defined	REFab/REFsb, FGR, leaky-bucket
ZQcal FSM	MC	Defined	MPC ZQCAL Start/Latch
RFM FSM	MC	Defined	RAA per bank, RAAIMT threshold
Power Mgmt FSM	MC	Defined	PD, SR, MPSM transitions
Write Data Path	MC	Defined	BL16 buffer, CRC, WL align, DFI write
Read Data Path	MC	Defined	Latency counter, CRC checker, ECS FSM
Rank Manager	MC	MISSING	rank_busy, rank_cke, rank_sr — all rank signals
Maintenance Arbiter	MC	MISSING	Collision resolution for REF/RFM/ZQ/MRR
Address Mapper	MC	MISSING	Physical addr decomposition
DFI Training Orch.	MC	MISSING	Training handshake boundary
DFI Adapter	MC	Weak	Signal encoding, freq ratio, latency

11 Verification Coverage Nodes Required

Document	Status	Coverage Model
Coverage_BankFSM.md	Not started	All 11 states, all transitions, illegal transition exclusions
Coverage_Refresh.md	Not started	REFab, REFsb, FGR, credit max, urgent stall, derating
Coverage_Turnaround.md	Not started	tWTR_L/S, tRTW, BG hit/miss combinations
Coverage_RFM.md	Not started	RAA increment, decrement, threshold hit, RFM injection
Coverage_DFI.md	Not started	All 6 DFI sub-interfaces, latency correctness
Coverage_Training.md	Not started	WL, read gate, DQ, VREF, CA, CS training sequences
Coverage_Powersave.md	Not started	PD entry/exit, SR entry/exit, MPSM, tXP, tXS, tDLLK

12 Time Estimate & Roadmap

12.1 Assumptions

1 person, part-time, ECE capstone level. Architecture first, RTL second.

12.2 Phase Estimates

Phase	Milestone	Time	Exit Criteria
Architecture Completion	ADR-0001 through ADR-0006 done	2–3 weeks	Architecture Freeze v1.0 signed off
Architecture (P1)	ADR-0007 through ADR-0010	1 week	Rank Mgr, Maint. Arbiter, Training, DFI owner defined
CIF RTL	AXI → CDC handoff complete	4–6 weeks	Simulation passing basic read/write burst
MC Core RTL	Init FSM through DFI adapter	6–10 weeks	All FSMs, timing gates, and scheduler RTL complete
DFI + PHY Interface	DFI adapter to PHY handshake	2–4 weeks	DFI training handshake functional
Training RTL	WL + read gate + DQ + VREF + CA	2–4 weeks	All training sub-sequences pass simulation
Verification	UVM TB, scoreboards, coverage	8–12 weeks	Coverage closure; DDR5 + DFI compliance
Total (architecture-first)		5–6 months	Aligns with cap-stone roadmap
Total (RTL without arch)		8–10 months	Multiple redesigns expected

12.3 Current Recommendation

Hold Architecture — Do Not Start RTL

Current architecture is approximately **40% complete** toward RTL readiness.

Complete the following before writing any RTL:

1. ADR-0001: Command Format
2. ADR-0002: Address Mapping
3. ADR-0003: CDC Contract
4. ADR-0004: Queue Architecture (resolve Option A vs B)
5. ADR-0005: Command Lifecycle
6. ADR-0006: Timing Ownership

After these six ADRs, architecture completeness will reach $\approx 60\%$, which is the minimum threshold to start CIF RTL safely.

12.4 Priority Upload Queue

Documents needed next, in priority order:

1. CIF architecture spec
2. AXI → command conversion detail
3. Address mapping scheme
4. DFI interface spec (all 6 sub-interfaces)
5. PHY architecture
6. Training architecture
7. Refresh / RFM design detail
8. Verification plan

phase. RTL not started. Next milestone: Architecture Freeze v1.0 after ADR-0001 through ADR-0006.